ENTERPRISE NETWORK SECURITY

Design, Implementation and Verification Report

Cisco Packet Tracer | Final project report | 29 October 2026

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Evidence basis: Cisco Packet Tracer screenshots supplied during project verification. Items not shown as tested are identified as pending.

1. Executive Summary

This project implements a segmented enterprise network for IT, Human Resources (HR), and Finance. A Cisco 2960 switch provides VLAN separation, a Cisco 2911 router routes between VLANs, and an ASA 5505 firewall provides perimeter routing, NAT/PAT, and HTTP/ICMP inspection. An extended access control list (ACL) blocks HR-initiated traffic to the Finance network. Port security on user-facing switch ports restricts each port to one learned MAC address.

Screenshots demonstrate successful internal routing and internet access from IT, an HR-to-Finance block, active 802.1Q trunking, sticky MAC learning, and firewall NAT counters. Some further tests are recommended before claiming full security validation.

2. Objectives and Scope

Objectives: (1) isolate departments with VLANs; (2) enable controlled inter-VLAN routing; (3) restrict HR-to-Finance access; (4) secure access switch ports; (5) translate internal addresses for internet access; (6) verify connectivity and document evidence.

Scope: Packet Tracer simulation. The project does not establish real-world production security, comprehensive firewall policy hardening, intrusion detection, or high availability.

3. Network Topology and Addressing

Observed logical path: IT-PC / HR-PC / FIN-PC → SW1 → R1 → FW1 → ISP-R1 → Internet-Server. The internet cloud icon appears in the topology, but its connection was not established by the supplied screenshots.

Node / VLAN	IP address / network	Role
IT / VLAN 10	192.168.10.0/24; gateway 192.168.10.1	IT departmental LAN
HR / VLAN 20	192.168.20.0/24; gateway 192.168.20.1	HR departmental LAN
Finance / VLAN 30	192.168.30.0/24; gateway 192.168.30.1	Finance departmental LAN
R1 ↔ FW1	10.0.0.1/30 ↔ 10.0.0.2/30	Internal transit link
FW1 ↔ ISP-R1	203.0.113.2/30 ↔ 203.0.113.1	External transit link
Internet-Server	198.51.100.10	Simulated internet service

Observed PC addresses: IT-PC 192.168.10.21, HR-PC 192.168.20.21, FIN-PC 192.168.30.21.

Figure 1 placeholder — Insert a full screenshot of the Packet Tracer logical topology with readable device names and connections.

4. Switch Configuration: VLANs and Trunk

SW1 uses VLAN 10 (IT) on Fa0/1, VLAN 20 (HR) on Fa0/2, and VLAN 30 (FINANCE) on Fa0/3. Gi0/1 is an active 802.1Q trunk allowing VLANs 10, 20, and 30; the observed native VLAN is 1.

```
SW1# show vlan brief
SW1# show interfaces trunk
```

Figure 2 — Capture the VLAN table showing Fa0/1–Fa0/3. Figure 3 — Capture the trunk status showing Gi0/1, 802.1Q, and VLANs 10,20,30.

5. Router-on-a-Stick and Static Routing

R1 uses subinterfaces Gi0/0.10, Gi0/0.20 and Gi0/0.30, tagged with dot1Q 10, 20 and 30. Their gateway IPs are .1 in each VLAN. Gi0/1 connects to FW1 at 10.0.0.1/30. R1 default route points to FW1 at 10.0.0.2.

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip access-group BLOCK_HR_FINANCE in
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

Figure 4 — R1 show ip interface brief. Figure 5 — R1 running configuration showing subinterfaces and default route.

6. Interdepartmental Access Control

An extended ACL named BLOCK_HR_FINANCE denies packets sourced from HR (192.168.20.0/24) destined for Finance (192.168.30.0/24), and permits other IP traffic. It is applied inbound on R1 Gi0/0.20.

```
ip access-list extended BLOCK_HR_FINANCE
 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip any any
```

Evidence: show access-lists displayed deny-rule match counts increasing from 8 to 20; HR-PC pinging 192.168.30.21 failed with “Destination host unreachable” from 192.168.20.1. IT-PC successfully pinged both HR and Finance. A Finance-to-HR ping also timed out in the supplied screenshot, but this is not a general bidirectional deny rule: return traffic to HR is affected by the inbound ACL on HR, while initiating traffic from Finance is not directly denied by the shown ACL. More protocol-specific tests are required to characterize all flows.

Figure 6 — R1 ACL entries and match counters. Figure 7 — HR-PC failed ping to FIN-PC. Figure 8 — IT-PC successful pings to HR-PC and FIN-PC.

7. Switch Port Security

Fa0/1–Fa0/3 have port security enabled, maximum one MAC address per port, sticky MAC learning, and violation mode restrict. The final secure address table displayed one sticky MAC for each VLAN/port: VLAN 10 Fa0/1 0060.4730.ACBA; VLAN 20 Fa0/2 0001.4351.1D25; VLAN 30 Fa0/3 0060.5C85.BCA1. The screenshots showed secure-up status and zero violations.

```
SW1# show port-security interface fastEthernet 0/1
```

```
SW1# show port-security address
```

Figure 9 — Port security details on Fa0/1. Figure 10 — Sticky MAC table for all three VLANs.

8. ASA Firewall: Interfaces, NAT and Inspection

FW1 inside VLAN1 is 10.0.0.2/30 with security level 100. Outside VLAN2 is 203.0.113.2/30 with security level 0. Static routes point to VLANs 10, 20, 30 via 10.0.0.1; default route points to ISP-R1 203.0.113.1.

```
object network INTERNAL_LAN
 subnet 192.168.0.0 255.255.0.0
 nat (inside,outside) dynamic interface
policy-map global_policy
 class inspection_default
  inspect http
  inspect icmp
service-policy global_policy global
```

FW1 show nat displayed translate_hits=170 and untranslate_hits=218, demonstrating NAT activity. show running-config confirmed HTTP and ICMP inspection are configured and the policy is applied globally. The Packet Tracer ASA rejected show service-policy; this command limitation is not itself evidence of a failed policy.

Figure 11 — FW1 interface status. Figure 12 — FW1 show route and show nat. Figure 13 — FW1 running configuration showing NAT and inspection policy.

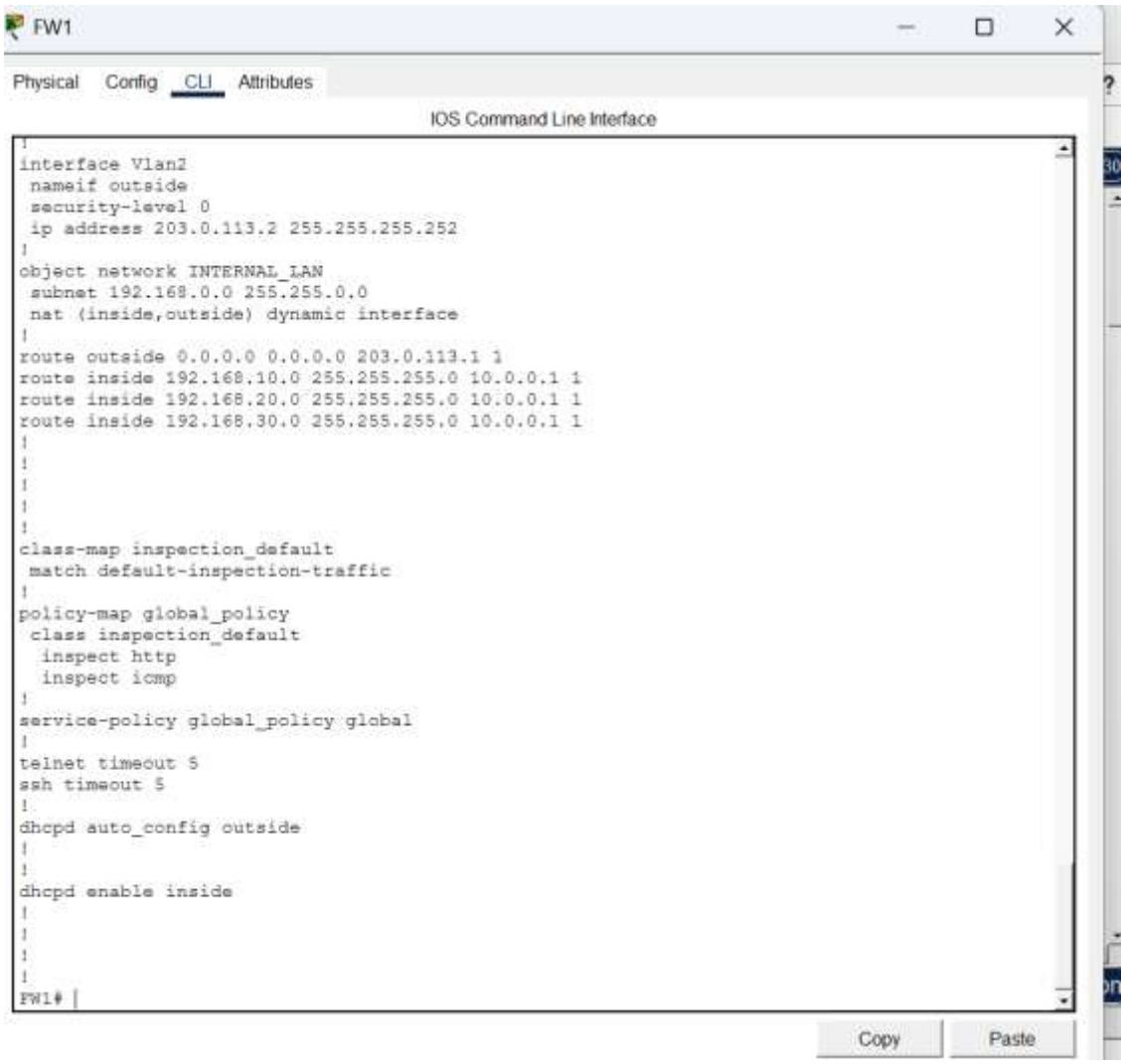


Figure 13. Supplied FW1 screenshot showing NAT, static routes, and HTTP/ICMP inspection.

9. Testing and Verification

Test	Observed result	Interpretation
IT → Internet-Server (198.51.100.10)	4 replies, 0% loss	Passed
IT → HR (192.168.20.21)	4 replies, 0% loss	Passed
IT → Finance (192.168.30.21)	4 replies, 0% loss	Passed
HR → IT (192.168.10.21)	4 replies, 0% loss	Passed
HR → Finance (192.168.30.21)	Destination host unreachable	Blocked as intended
Finance → HR (192.168.20.21)	Request timed out	Observed failure; interpret with ACL direction
FIN → IT gateway (192.168.10.1)	4 replies, 0% loss	Passed
Switch sticky MAC learning	3 addresses learned	Passed
FW1 NAT activity	170 translation / 218 untranslation hits	Active

Unauthorized host on protected switch port	Not tested in evidence	Pending
Full HTTP policy traffic inspection	Configured; browser loaded server page	Not independently proven by packet trace

Figure 14 — FIN-PC blocked HR ping. Figure 15 — IT-PC browser successfully displaying the simulated internet server at http://198.51.100.10. For every figure, retain the PC/device name, command, destination, and result in the same screenshot.

10. Security Evaluation and Limitations

Strengths: VLAN segmentation reduces broadcast domains; an ACL enforces a specific HR-to-Finance restriction; sticky MAC port security limits new devices on access ports; NAT/PAT hides internal addresses from the simulated outside network; firewall inspection is configured for HTTP and ICMP.

Limitations: The ACL shown does not constitute a complete least-privilege departmental policy. The native trunk VLAN remains VLAN 1. No evidence was supplied for disabling unused ports, SSH-only management, administrative authentication hardening, logging, DHCP snooping, DAI, or a port-security violation test. The inside-to-outside default firewall policy and application-level access controls require additional review for production use.

11. Recommended Final Checks

1. Save configurations on SW1 and R1 with copy running-config startup-config and on FW1 with write memory; confirm prompts complete. 2. Save the .pkt project. 3. Retest the six ping scenarios and capture the outputs. 4. Verify a deliberately unauthorized MAC triggers port-security restrict without disrupting your final project; reset/restore if needed. 5. Confirm HTTP access from each department as permitted by the intended security policy. 6. Review whether Finance-to-HR access should be allowed or blocked and document the expected direction.

12. Conclusion

The Cisco Packet Tracer enterprise network demonstrates working VLAN segmentation, router-on-a-stick inter-VLAN routing, HR-to-Finance ACL enforcement, sticky MAC port security, firewall routing, dynamic NAT/PAT, and configured HTTP/ICMP inspection. Available tests show IT access to both departments and to the simulated internet server. The project meets the main demonstrated implementation goals, while the outstanding validation and hardening tasks should be completed before presenting it as production-ready.

Appendix A — Screenshot Checklist

Figure 1: Full logical topology — [insert or verify screenshot]

Figure 2: SW1 VLAN assignments — [insert or verify screenshot]

Figure 3: SW1 trunk status — [insert or verify screenshot]

Figure 4: R1 interface status — [insert or verify screenshot]

Figure 5: R1 subinterfaces and default route — [insert or verify screenshot]

Figure 6: R1 ACL and counters — [insert or verify screenshot]

Figure 7: HR blocked from Finance — [insert or verify screenshot]

Figure 8: IT successful departmental pings — [insert or verify screenshot]

Figure 9: SW1 port security status — [insert or verify screenshot]

Figure 10: SW1 sticky MAC table — [insert or verify screenshot]

Figure 11: FW1 interface status — [insert or verify screenshot]

Figure 12: FW1 routing and NAT counters — [insert or verify screenshot]

Figure 13: FW1 inspection running configuration — [insert or verify screenshot]

Figure 14: FIN-to-HR test — [insert or verify screenshot]

Figure 15: Internet browser page — [insert or verify screenshot]

Appendix A — Screenshot Evidence

The following screenshots were supplied during the Cisco Packet Tracer verification process. Each is included as visual evidence. Screen captures show the state at the time of capture; they do not replace any outstanding tests identified in the report.

Figure A1. Network topology

Topology with departmental PCs, SW1, R1, FW1, ISP router and server.

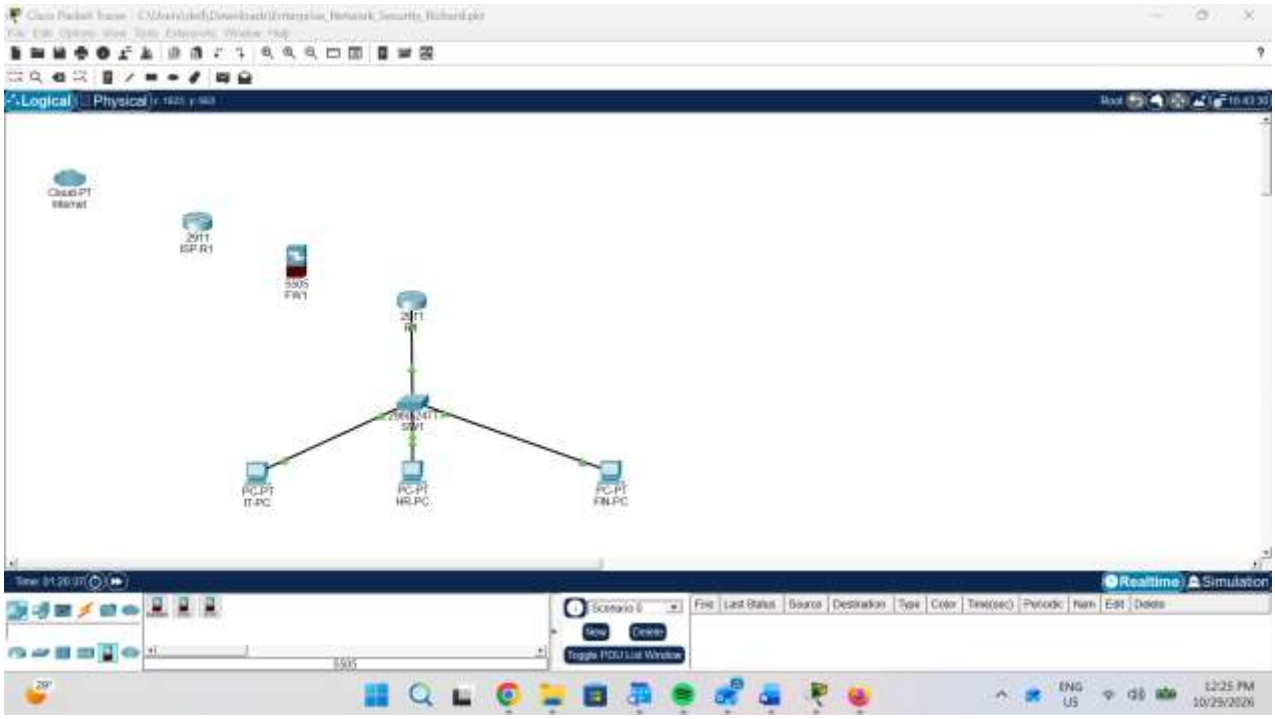


Figure A2. SW1 VLAN membership

VLAN 10 IT, VLAN 20 HR, VLAN 30 Finance and access ports.

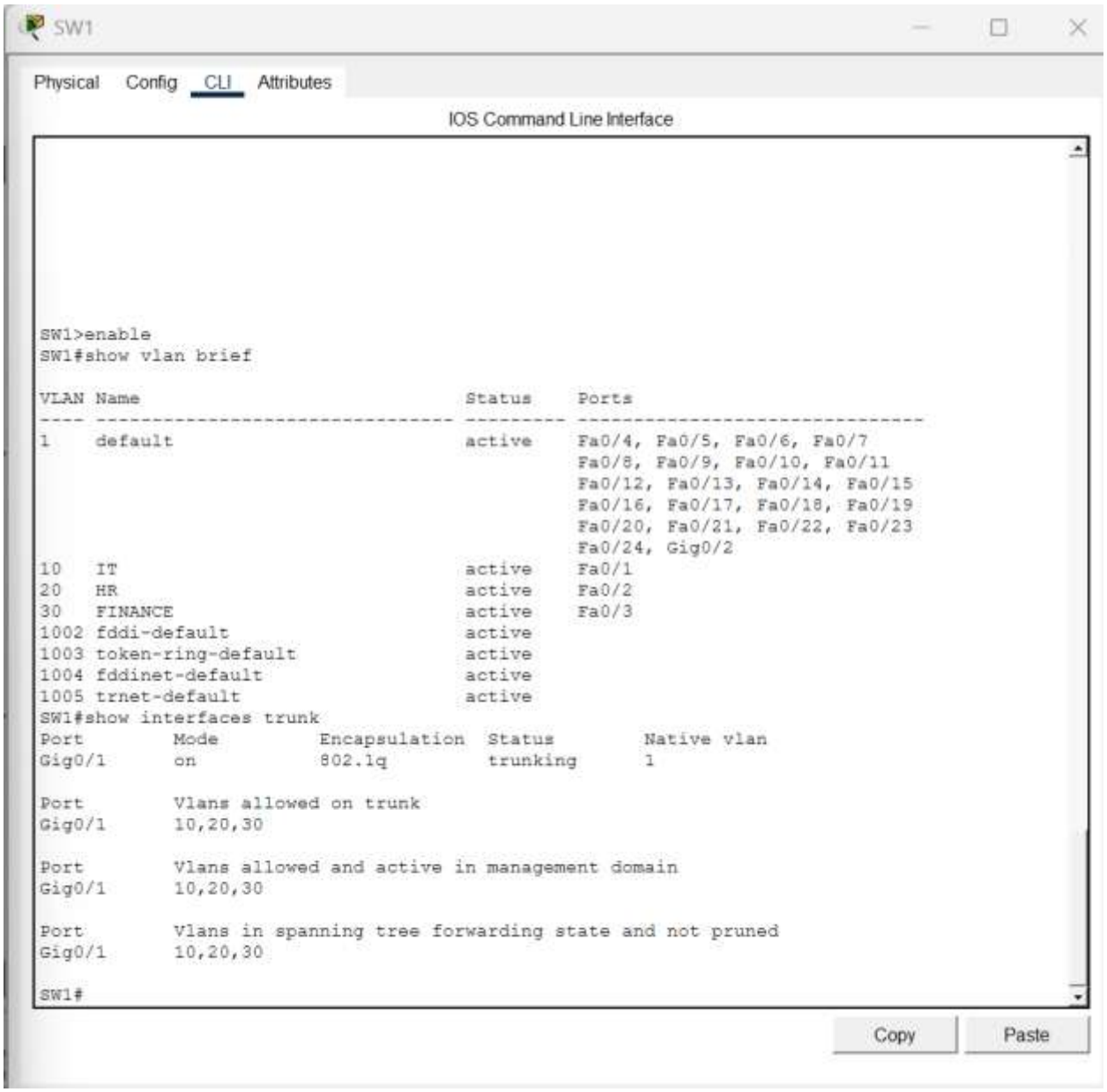


Figure A3. SW1 trunk verification

Gig0/1 trunking with VLANs 10, 20 and 30 allowed.

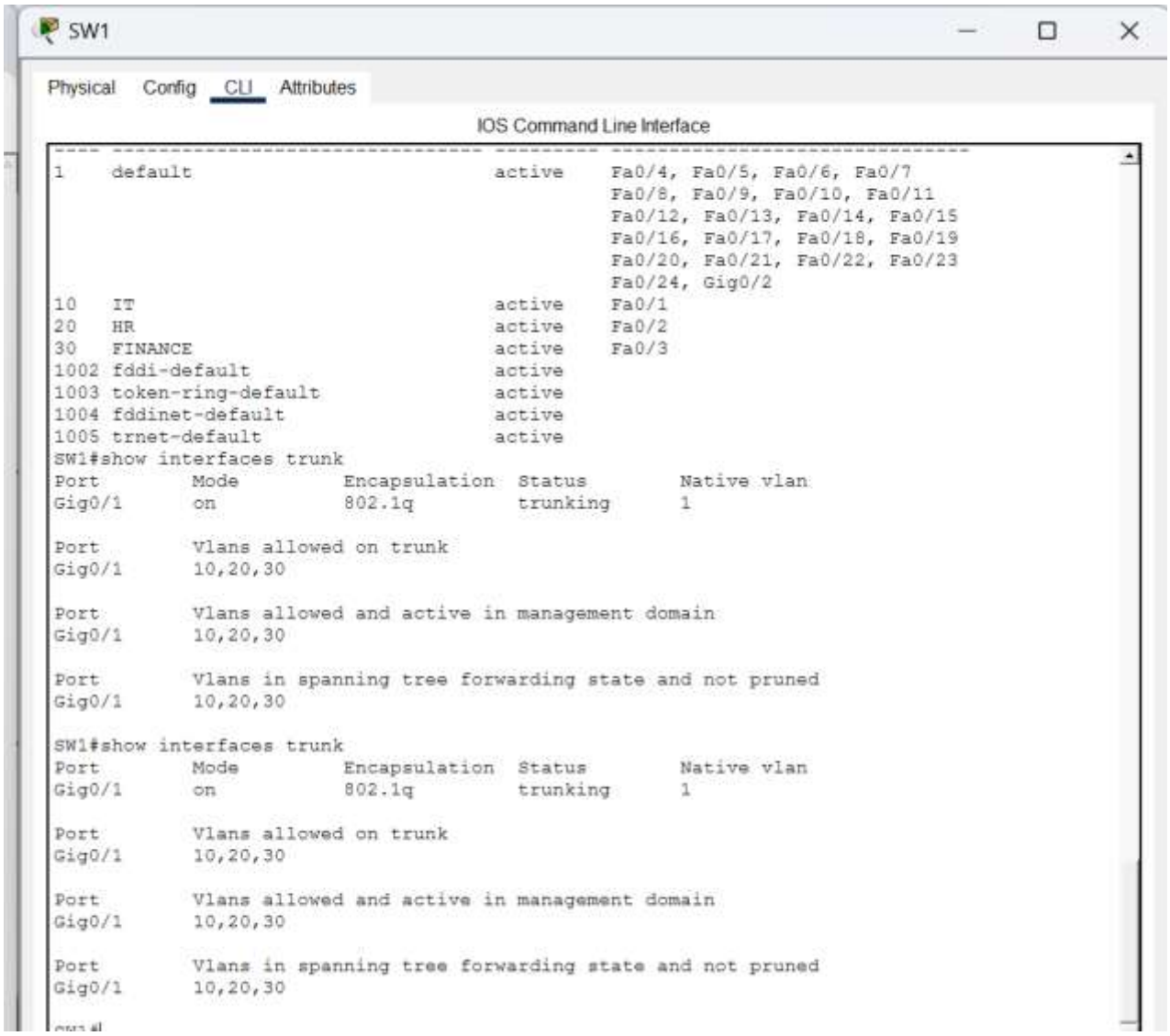


Figure A4. SW1 port security configuration

Port security enabled, restrict mode and maximum one MAC address.

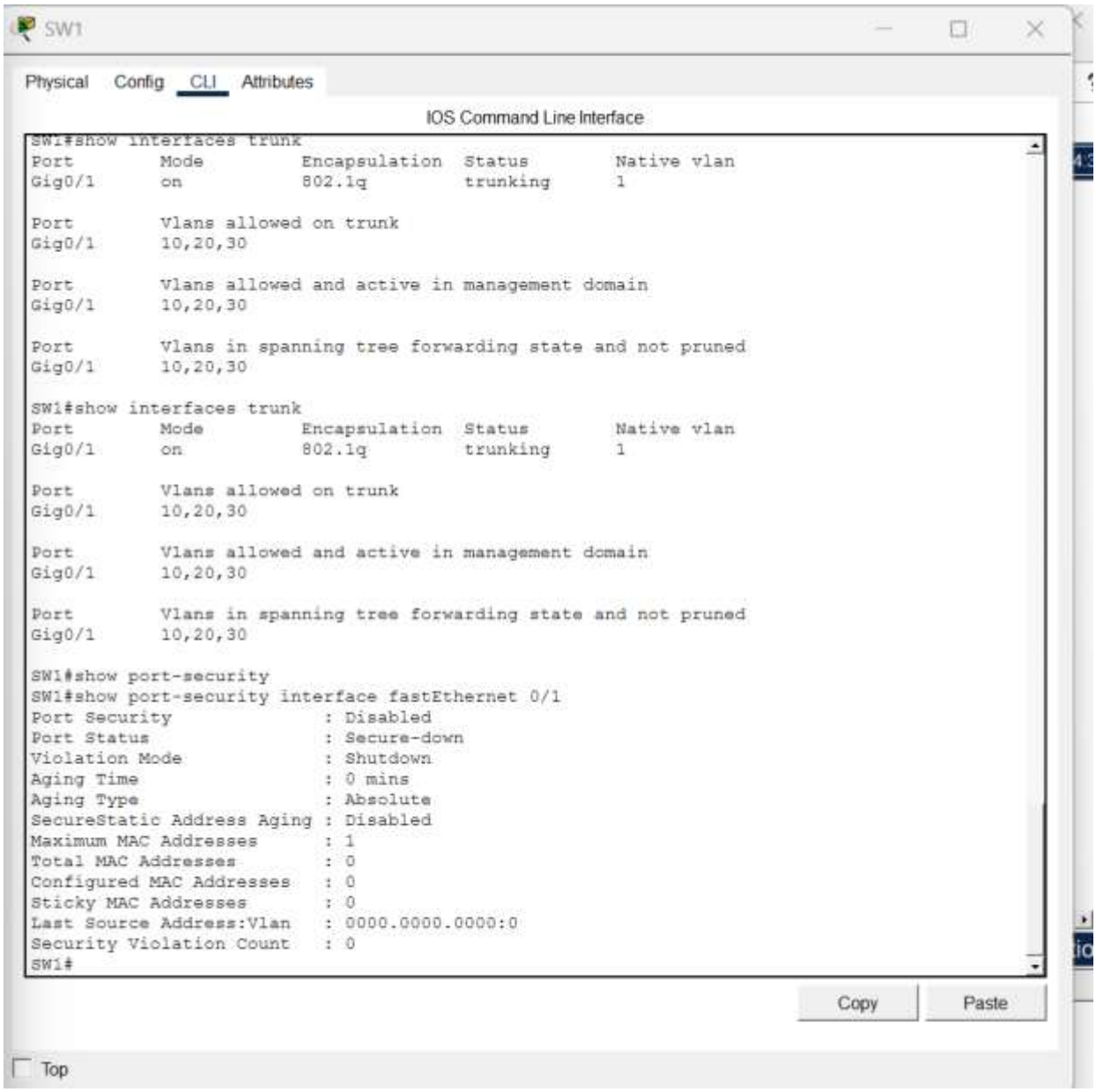


Figure A5. SW1 sticky MAC verification

Learned secure sticky MAC addresses for departmental access ports.

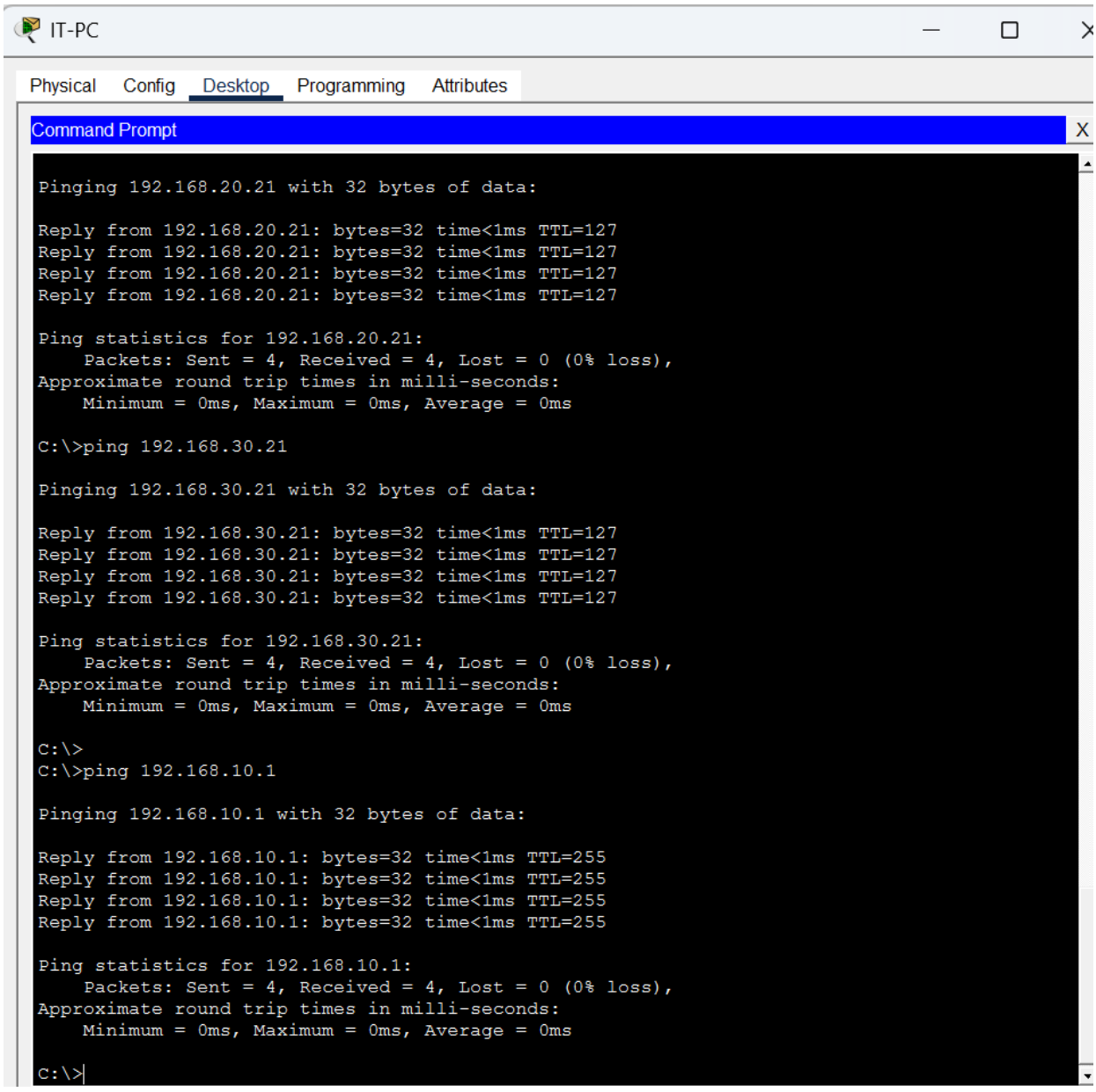


Figure A6. R1 interface addressing

Router subinterfaces and gateway IP addresses.

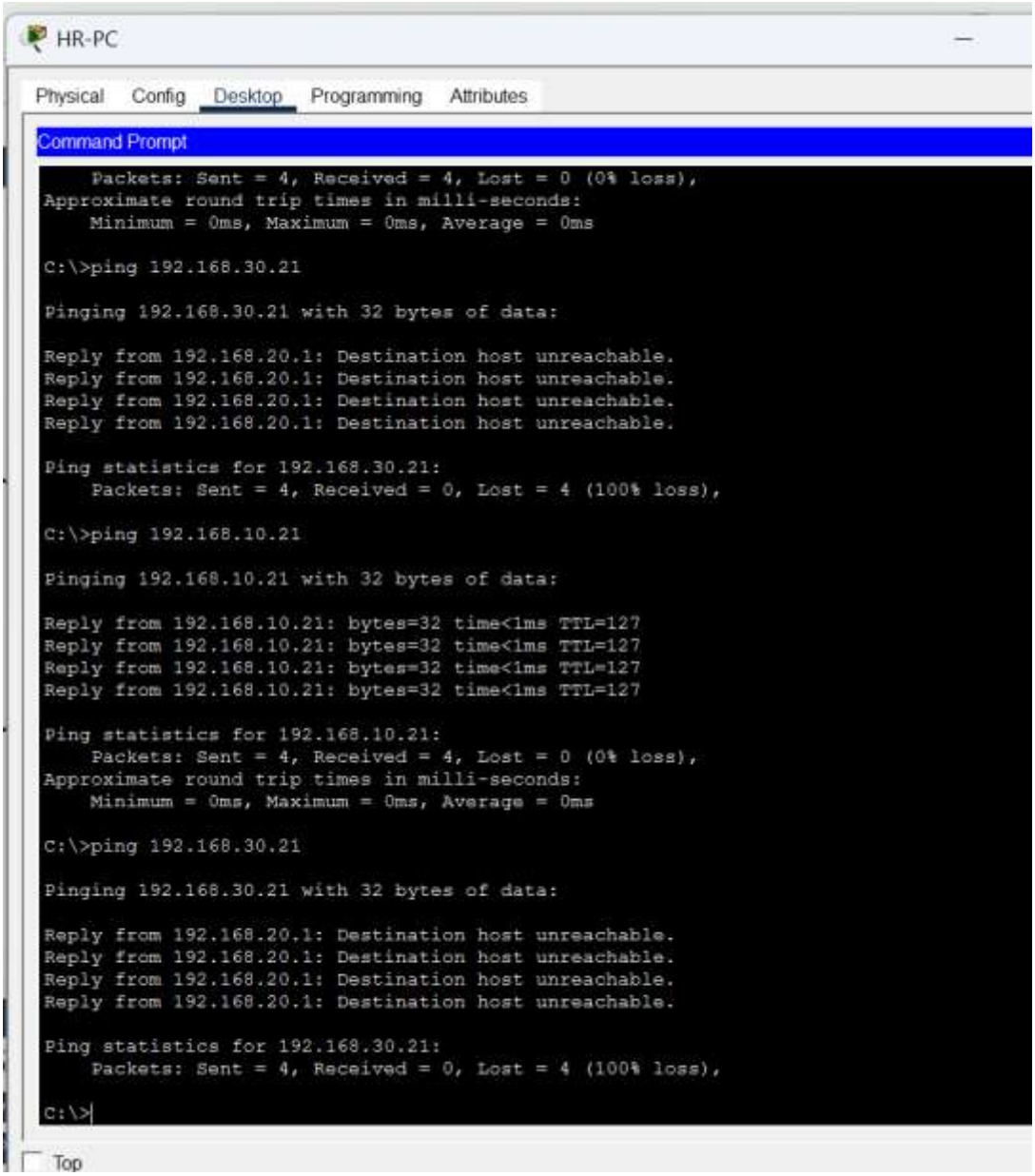


Figure A7. R1 ACL configuration

HR-to-Finance deny rule and permit rule.

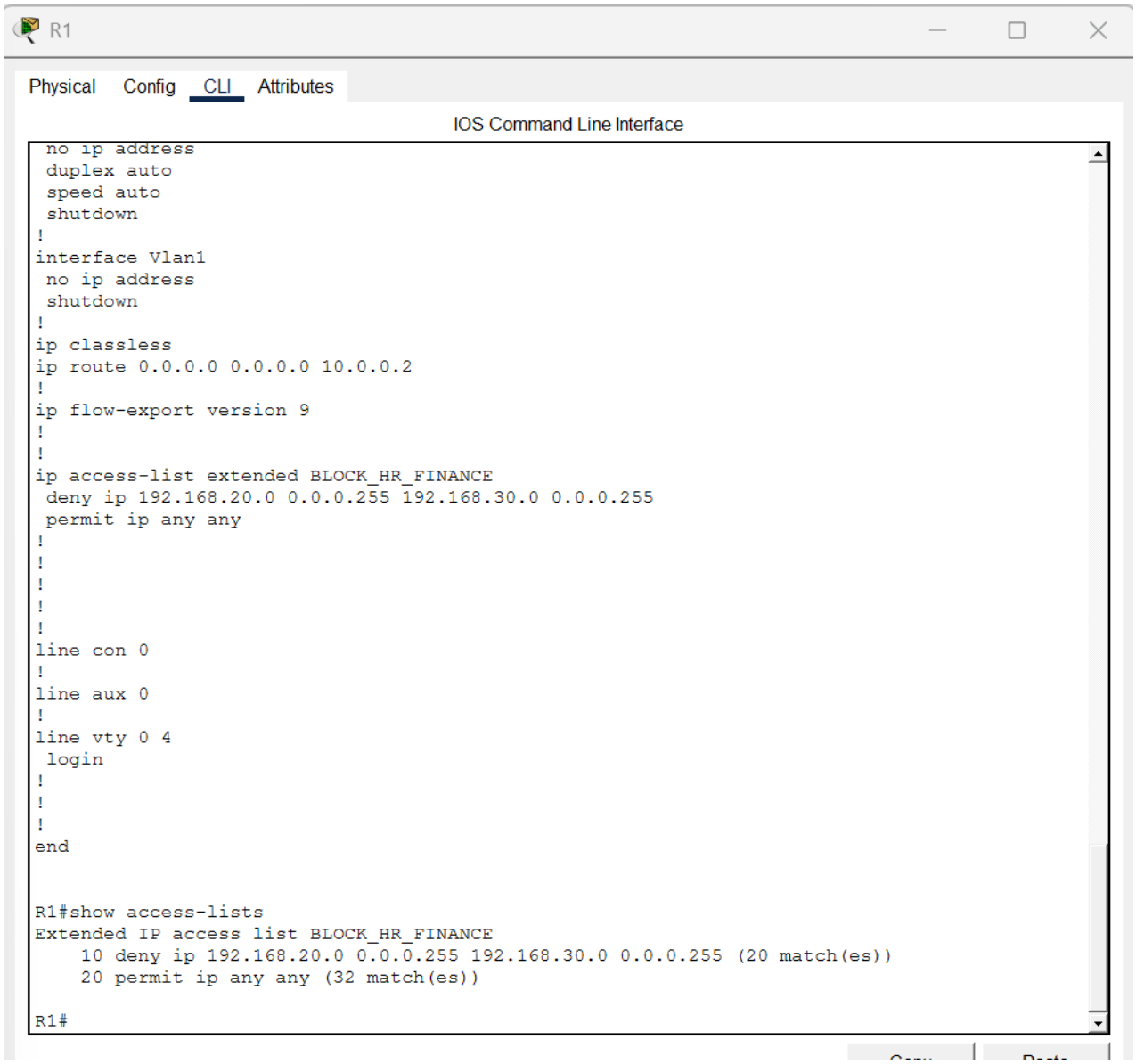


Figure A8. HR-to-Finance connectivity test

HR cannot reach Finance, while IT reachability is shown.

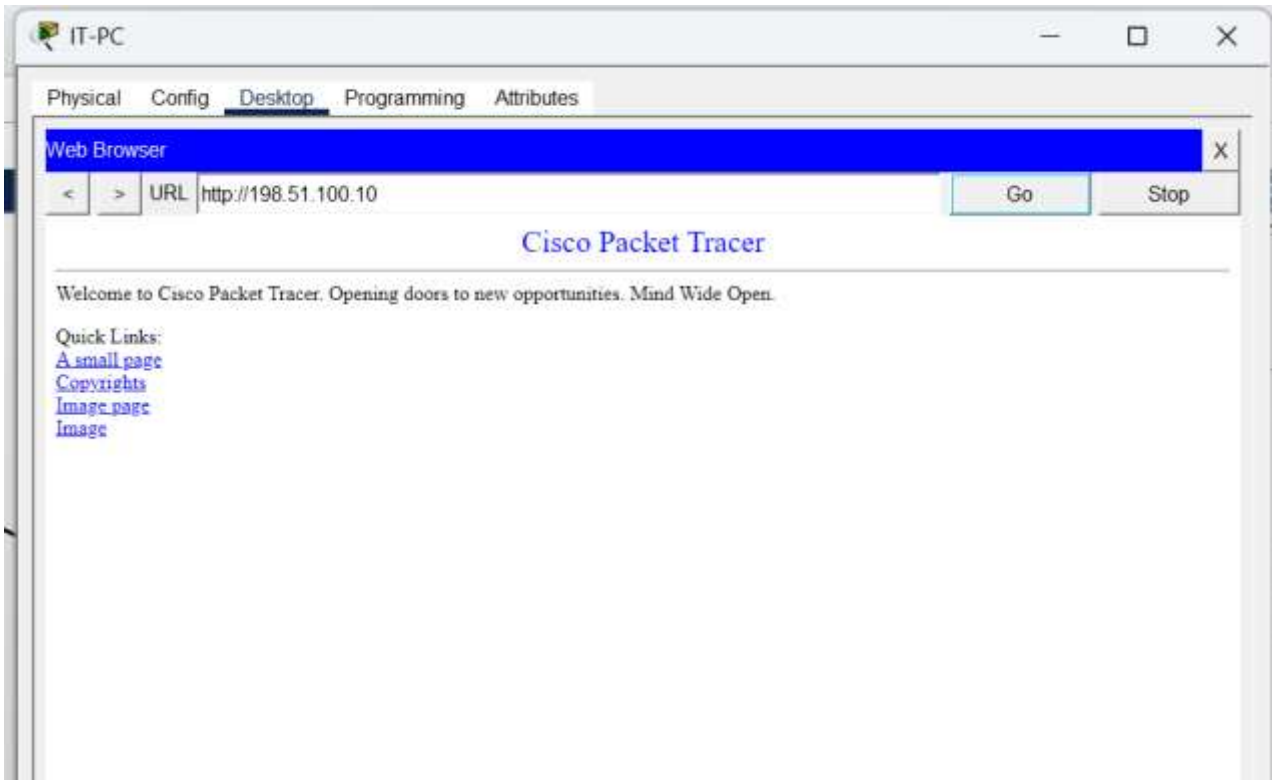


Figure A9. IT connectivity testing

IT PC ping results showing inter-VLAN access.

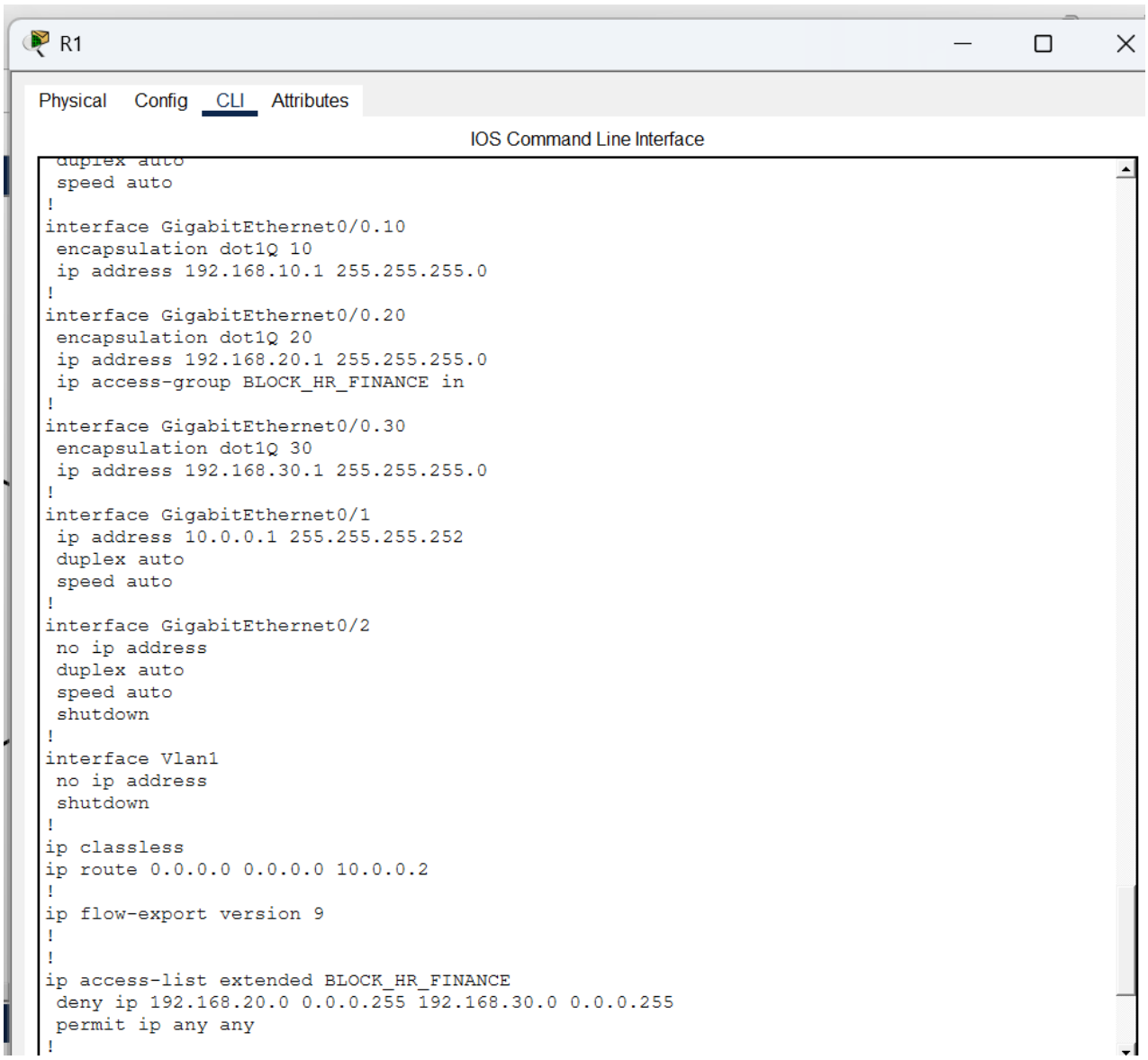


Figure A10. FW1 interface verification

Inside and outside firewall interfaces.

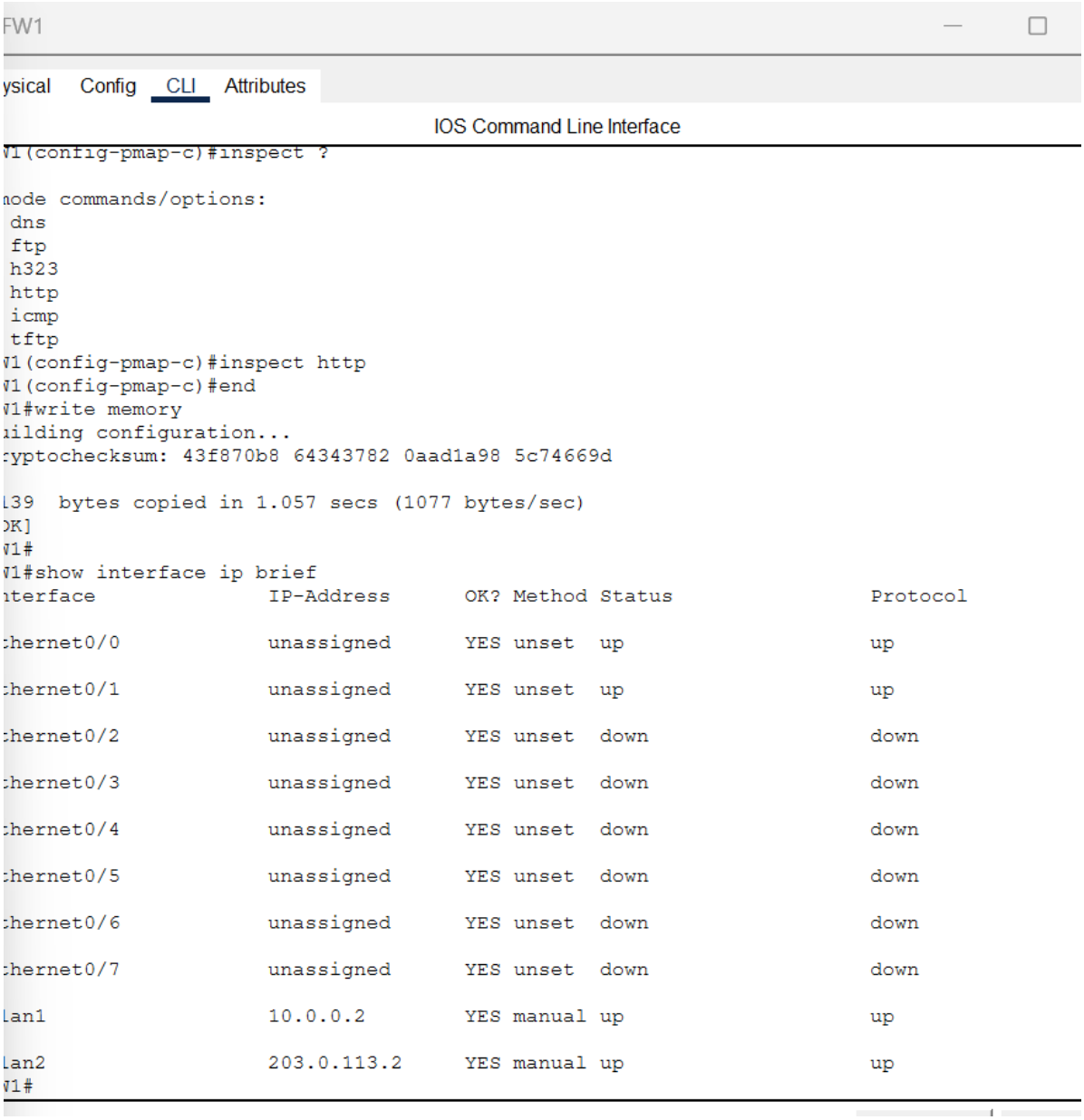


Figure A11. FW1 static routing

Routes toward departmental VLANs and default route.

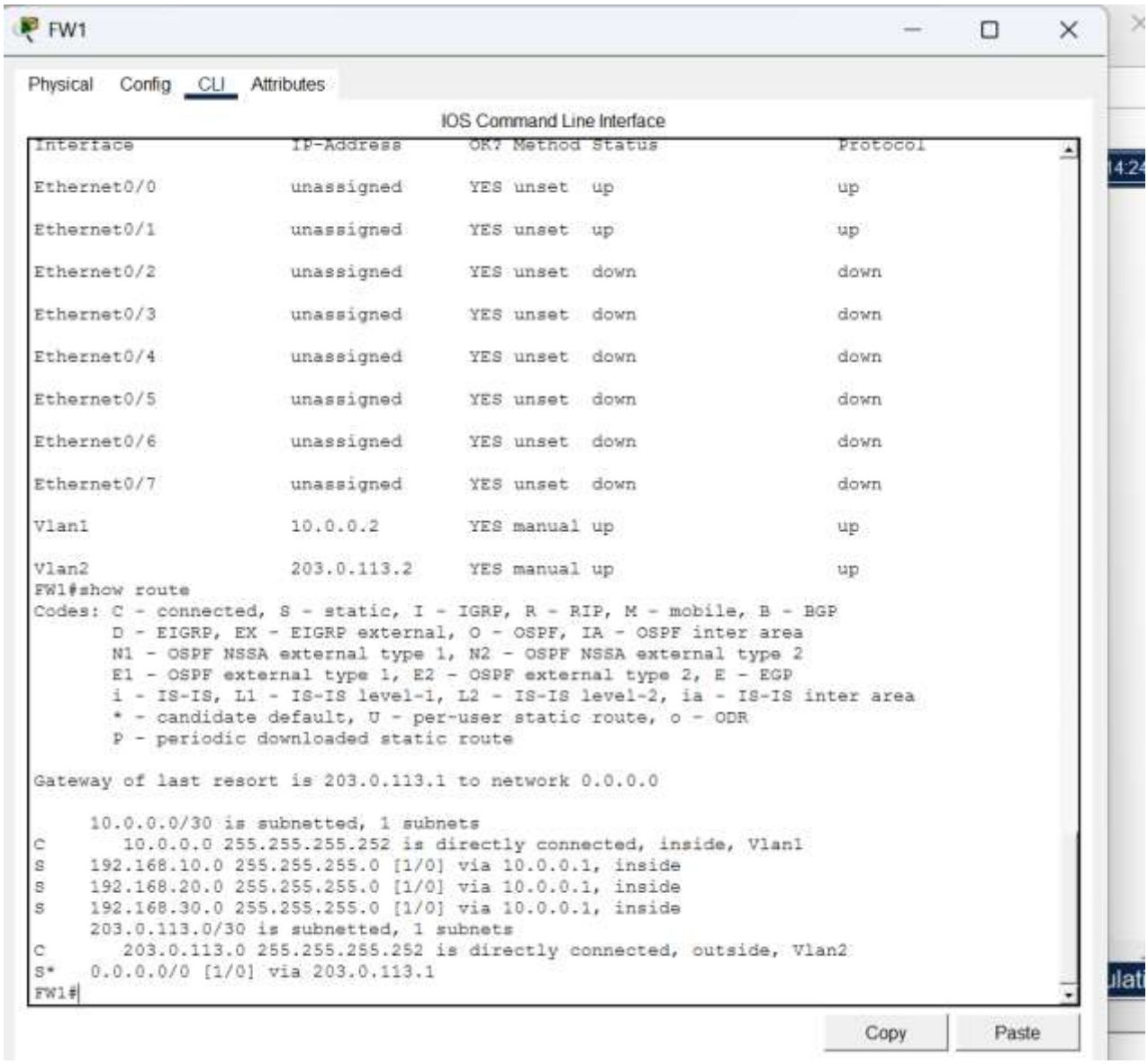


Figure A12. FW1 NAT statistics

NAT policy and translation hit counters.

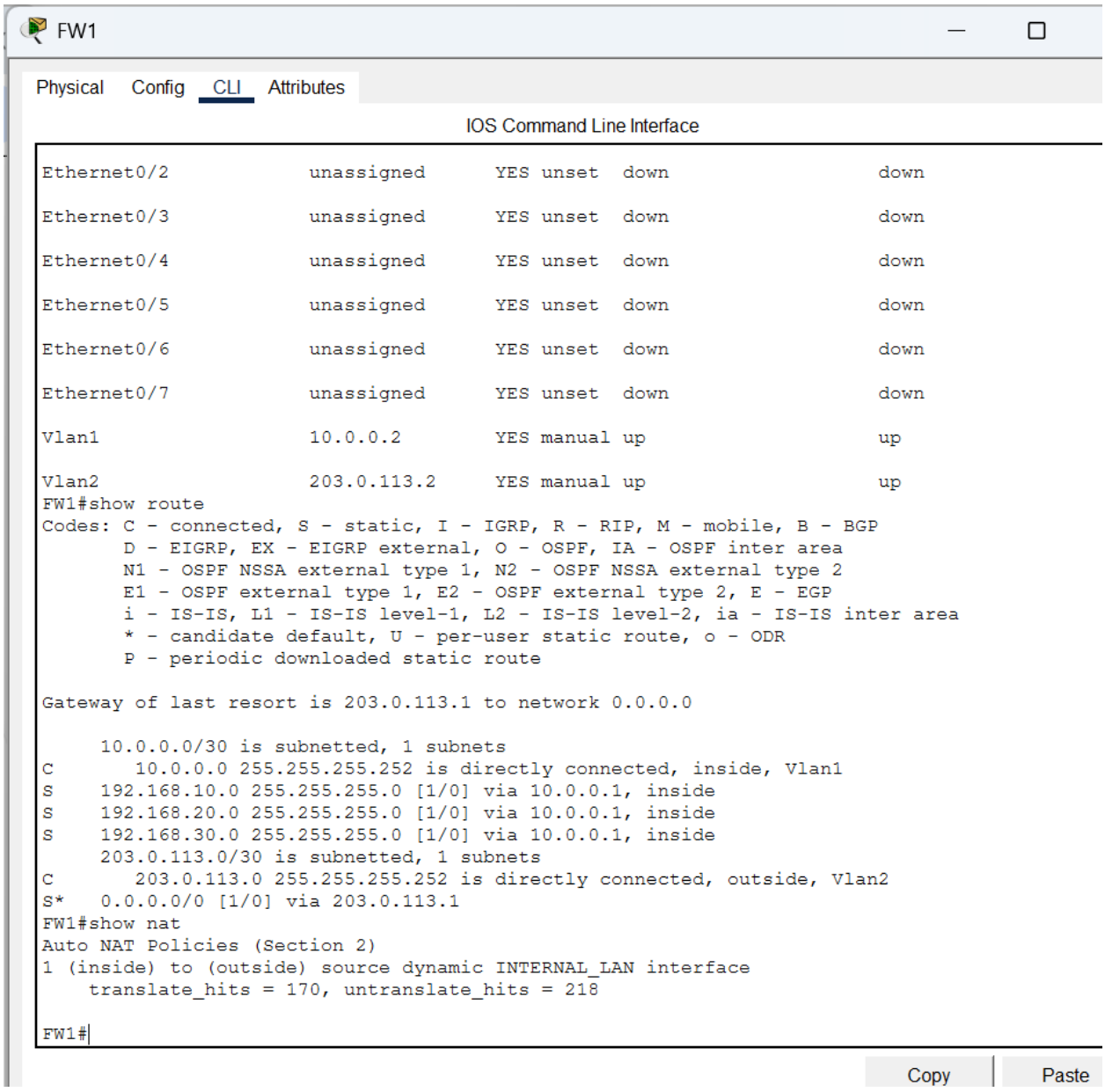


Figure A13. FW1 inspection policy

HTTP and ICMP inspection and applied global service policy.

